# JESICA MARIA RAMIREZ TOSCANO

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Applied Data Scientist with **7+ years** of experience in **statistical data analysis**, **experimental design** and **machine learning** for marketplace-quality and regulatory decisions. Focused on measuring whether products, policies, and AI systems work, especially in the public interest.

## PROFESSIONAL EXPERIENCE

### *SCIENTIST II, UBER*

***Oct 2024 – Present · Earner Progression Science***

- **Designed experiments and behavioral measurement frameworks** across Product, Engineering, and Operations; contributed to a real-time feedback system generating **$36M in annual cost savings**.
- **Built a causal estimation framework** using order-level fixed-effects regression to quantify the marketplace cost of courier behavior, enabling economically grounded quality measurement and informing policy decisions on earner-quality metrics.
- **Narrowed earner-quality interventions with uplift modeling,** identifying the smaller set of earners worth targeting, and analyzed earner-quality metric threshold changes to improve marketplace outcomes.
- **Data lead for L4–L5 incidents in Earner Metrics:** supported incident response across the earner organization, helping engineers validate with data that live changes solve any user gaming.

### *APPLIED SCIENTIST II, UBER*

***Sep 2021 – Oct 2024 · Policy Research – Legal Data & Economics***

- **ML and LLM classification for regulatory compliance:** replaced manual and keyword-search review of incidents for Australian regulators (1% precision, 50% recall) with a logistic-regression text classifier (17% precision, 89% recall; **projected savings of AUD 500,000 per year**), then with an LLM-based classifier behind a PII-redaction layer (24% precision, 91% recall). Nominated for the 2023 Reimagine Award.
- **Evidence for regulators:** developed multi-scenario economic analyses to support regulatory engagement with the **New York Attorney General**, translating complex platform data into externally auditable evidence for a government proceeding.
- **Built self-serve tools for Legal:** a scenario simulator that lets Legal directors estimate litigation exposure under different assumptions, and a driver-activity dashboard that **10+ paralegals** use to self-serve a driver's history and activity over any period.
- **Regulatory data productions:** built a privacy framework for datasets produced to regulators and external parties; modeled litigation risk and financial exposure.

### *DATA SCIENTIST, DEEP DIVE*

***Jun 2020 – Aug 2020***

- Helped build the company's **NLP pipeline** for large-scale text data.

### *ECONOMIST, BANCO DE MÉXICO (CENTRAL BANK)*

***Aug 2017 – Aug 2019 · General Directorate of Economic Research***

- **Built an NLP policy-tone index** across **80+ central-bank statements** (2008–2019) and derived inflation expectations and interest-rate forecasts (1 month to 3 years) from swaps, bonds, and surveys, supporting monetary policy decisions.

## EDUCATION

**University of Chicago** *Sep 2019 – Jun 2021*

*M.S. in Computational Analysis and Public Policy, With Honors*

Joint degree by the Department of Computer Science and the Harris School of Public Policy. Coursework: Computer Science with Applications I & II, Databases, Cloud Computing, Linear Algebra, Machine Learning, Advanced Machine Learning, Data Visualization, Advanced Statistics, Causal Inference.

**Universidad de las Américas Puebla, Mexico** *Aug 2012 – Jun 2017*

*B.A. in Economics, Magna Cum Laude*

First Place, Empirical Research Thesis Regional Competition in Puebla

## SKILLS

**Programming Languages:** Python, SQL, R, C#, JavaScript

**Data and Dev Tools:** Presto, Spark, Docker **ML Tools:** PyTorch, scikit-learn, transformers

## PROJECTS

- **Classification and Prediction:** **COVID-19 in Mexico**: predicting severe disease outcomes. Used balanced and weighted Random Forest models to predict individual deaths and hospitalizations from government data.
- **Neural Networks and Social Polarization:** **Predicting Polarization from News Outlets Tweets**. Built a convolutional neural network using PyTorch and trained the model to learn the social polarization of the comments given a tweet posted by a news media outlet. This research aims to contribute to creating a more civil and less polarized space for discourse on social media.
- **Full Web Stack Development:** **TenisLAT**: built a bilingual (Spanish/English), mobile-first web app for level-based amateur tennis leagues in Mexico City and La Paz. Players self-assess a level, enroll in a season, report match scores, and climb a per-level ranking driven by a Glicko-2 rating engine. Built with TypeScript, Next.js, Prisma, and PostgreSQL.
- **Databases and App Management:** **Mapping Crimes at a Certain Day and Hour in Mexico City**. Built a web app to provide information on the average number of crimes that happened at a specified place in Mexico City using Google's API and **2M+ observations** from the City Government's API.
- **Web Scraping and Time-Series Prediction:** **Hohonu: Water Level Monitoring**. Given the lack of quality and standardized water level data, collaborated on the pipeline that cleans, calibrates, and predicts local water level data. Scraped data from **200+ URLs** to obtain the standard measures (datums) needed for calibration, and optimized the previous VAR model predictions with additional external data.